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MODULE 2

MATERIALS FOR ELECTRONIC APPLICATIONS

- Nanomaterials are materials whose characteristic length scale lies within the Nano metric range (1-100nm) at least in one dimension.
- Nanoparticles that are naturally occurring (e.g. volcanic ash, soot from forest fires) or are the incidental by-products of combustion processes (e.g. welding, diesel engines)
- Nanomaterials have a much greater surface area per unit mass compared with larger particles. Shows enhance properties such as reactivity, strength and electrical characteristics

Classification

1. Classification based on dimension
2. Classification based on materials

Classification based on dimension

a. **Zero dimensions (0-D):** Here all the three dimensions are in the nanometric range eg. Nano particle.
b. **One dimension (1-D):** Here one of the dimensions is outside the nanometric range and the other two are within the range eg. Nano wires, fibers and tubes.
c. **Two dimension (2-D):** Two of the dimensions are outside the Nano metric range and one is within the range eg. Nano films
d. **Three dimension (3-D):** All the dimensions are outside the nanometric range(>100nm) eg. Bundles of Nano wires and tubes.

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Classification based on materials

(a) **Carbon based nanomaterials**

- Materials in which the nano component is pure carbon eg. Carbon nano tubes (CNT), wires, spheres (fullerene) and graphene

(b) Metal based nanomaterials

- Materials made of metallic nanoparticles like gold, silver, metal oxides, etc. eg. Titanium dioxide (TiO_2).

c) Nano composites

- It is a mixture of dissimilar nanoparticles or compounds (1D, 2D, 3D etc.) such as nano sized clays within a bulk material.
- They give better physical, mechanical and chemical properties to that of the initial bulk material.

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(d) Nano polymers or Dendrimers

Dendrimers are nano sized polymers built from branched units. These are tree-like molecules with defined cavities and can hide molecules in their cavities. Important application is for drug delivery.

(e) Biological nanomaterials

These nanomaterials are of biological origin and are used for nano technological applications. eg. DNA nano particles, nanostructured peptides and are used in drug delivery systems

Synthesis of Nanoparticles

The chemical methods are

a. Hydrolysis b. Reduction

<u>Hydrolysis</u>

Nano particles of metal oxides can be prepared by the hydrolysis of their alkoxide solutions under controlled conditions. Commercially important nano particles of silica (SiO_2), titania (TiO_2),), alumina (Al_2O_3) are prepared by this method.

$$\mathbf{Ti(OR)_4 + 2H_20 \Rightarrow TiO_2 + 4ROH}$$

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Sol-Gel method

- The sol-gel method is based on the phase transformation of a sol into a gel.
- A sol is colloidal system of nano-solid particles dispersed in a liquid.
- A gel is colloidal system in which liquid droplets are dispersed in a network of solid nanoparticles.
- The sol contains many other impurities. In order to remove impurities sol is transformed into a gel by changing the pH or other factors.
- The gel can be purified by filtration and washing with suitable solvents. The purified gel on drying gives solid nano particle.
- For example aluminium oxide nano particles are obtained by hydrolysis of aluminium alkoxide by sol gel technique.

$$Al(OR)_3 + 3H_2O \rightarrow Al(OH)_3 + 3ROH$$

Reduction

Nano particles of gold and silver can be prepared by the reduction method

Reduction using reducing agents

- Silver nanoparticles can be prepared by this method.

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- About 60 mL $AgNO_3$ solution is taken in a beaker, covered with a watch glass and heated in hot plate, then stirred using a magnetic stirrer. Add 6mL of trisodium citrate drop wise to the boiling solution.
- The beaker is then closed and kept for some time till the colour of the solution changed to a light golden colour.
- Then it is allowed to cool and the solvent can be removed by freeze-drying.

Applications

- Nano structured semiconductors are used as window layers in solar cells.
- Nano sized metal-oxides are used as gas sensors.
- Carbon nano tube are used for making paper batteries
- A mixture of Carbon nano tube and fullerenes is used for making solar cells.
- Nano cadmium telluride exhibit different colours and can be used dyeing fabrics.

Graphene

- Graphene is an allotrope of carbon consisting of mono layer of carbon atoms arranged in a two dimensional honey comb lattice, which is the building block of graphite materials. Eg. Carbon nano tube, fullerene.
- It has charge transport, thermal, optical and mechanical properties. Different techniques such as electron microscope and optical microscope etc, are used to characterisation of graphene

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Structure:

- In graphene the carbon atoms are Sp2 hybridized and are arranged in hexagonal fashion.
- Graphene is two dimensional planar structures consisting of single layer of carbon atoms that are tightly bound together by strong covalent bonds with bond length 0.142nm.
- Each carbon atom is connected to its three nearest neighbours by σ-bonds, and a delocalized bond ($\pi - \pi *$)
- Free moving electrons are responsible for electronic properties of graphene
- Since graphene is flat, every atom is on the surface and is accessible from both sides, so there is more interaction with surrounding molecules.

Properties

- Electrical conductivity: Graphene has high electrical conductivity
- Mechanical strength and elasticity: Graphene is 200 times stronger than steel. Graphene have great stability and a very high tensile strength (the force in which you can stretch something before it breaks).
- Optical: Graphene is suitable for transparent electrodes for displays, screens etc.

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- Surface area: It has high surface area due to its single-atom thickness and applications in energy storage devices

Applications

1. Electronics: Graphene based thin film can be used in touchscreens. Smaller size transistors can be developed using graphene
2. Energy storage: Graphene incorporated lithium-ion batteries have longer life span, faster charging time and higher capacity. Graphene is used in supercapacitors to provide high energy density.
3. Biomedical: Used to carry chemotherapy drugs to cancer cells and biosensors
4. Environmental: Graphene based membranes can purify water in a more efficient
5. It can be used in filters and coatings to remove pollutants and toxins from the air.

Carbon Nanotubes

Carbon nanotubes are tubular form of Carbon atom with diameter 1-3 nm and length few nm micro meters. CNT and made by rolling a one atom thick layer of graphite called graphene into a cylinder.

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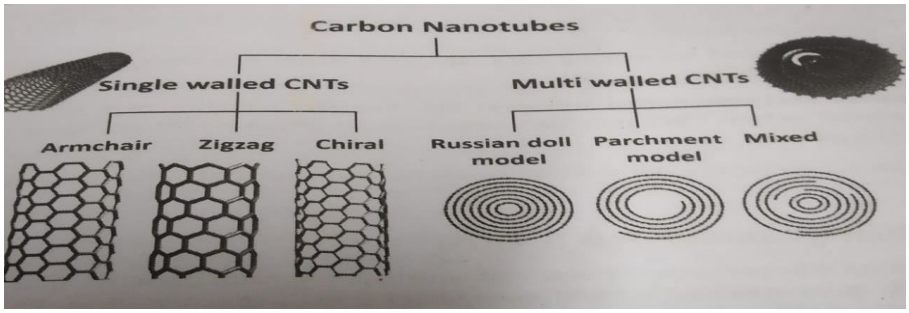
There are two types of CNTS. Single walled Single walled and multiwall nanotubes

Single walled nanotubes (SWNT)

SWNTs have a diameter close 1nm and tube length that can be many millions of time longer. The SWNT is formed by wrapping a one atom thick layer of graphite called graphene into seamless cylinder.

Multiwall carbon nanotubes (MWNT)

It consists of several multiple concentric nanotube cylinders, It Shows both metallic and semiconducting properties. It is used for storing fuel such as hydrogen and methane.



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Properties

1. Carbon nanotubes are very strong and have high thermal conductivity.
2. Due to the presence of delocalized electron they possess high electrical conductivity compared to metals.
3. Their tensile strength is 100 times greater than that of steel of same diameter.
4. They are light weight and density is about 1/4 of steel.

Applications

1. In the field of biomedicine CNT are used as capsules for drug delivery, biosensing, bioimaging, nanorobotics, gene therapy and tissue regeneration
2. Depending on their size they act as electrical condctors, semiconductors or capacitors.
3. It is used in battery technology and in industries as catalyst.
4. Used in solar panels, LEDs, sensors, transistors, field emitting devices, fuel cells etc.
5. It is used in water softening process as a filler
6. They are used as tips for analysis of DNA and Protein's by atomic force microscopy.

Fullerenes

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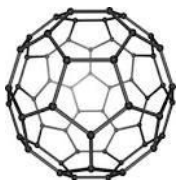
- A fullerene is an allotrope of carbon whose molecules consist of carbon atoms connected by single and double bonds so as to form a closed or partially closed mesh, with fused rings of five to six atoms. Fullerene is zero dimensional, hollow, closed cage nanoparticles.
- The most common fullerene molecule is having a molecular formula C60



Preparation

- Fullerenes are prepared by vaporizing a graphite rod in helium atmosphere. Mixture of fullerene like C60, C70 etc. is formed, which are separated by solvent extraction. Pure C60 is isolated from this mixture by column Chromatography.

Structure

- The C60 molecule has a perfect Icosahedron geometry. An icosahedron is a polygon with 60 vertices and 32 faces, 12 of which are pentagonal and 20 hexagonal. A carbon atom is present at each vertex of the structure. The molecule is aromatic and has several resonance structures.

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- The valencies of each carbon atom are satisfied by two Single and a double bond. C 60 is also known as bulky ball as it is a spherical cluster of carbon atoms arranged 5 and 6 member rings ball shape form a of soccer
- Carbon atom in fullerene molecule is SP^2 & SP^3 hybridized

Properties

1. Fullerene is black powdered material and forms deep magenta solution when dissolved in benzene.
2. Fullerene became harder than steel and diamond. Very tough and thermally stable and exists as a discrete molecule, unlike other allotropic forms of carbon
3. Electrical: They are normally insulators; It can be used as a super conductor when doped with alkali metals.
4. They give absorption in UV region**.**

Applications

- **Coating &Lubricant**: It is suitable for use as a lubricant due to its spherical structure. The bulky balls, would act as a molecular ball bearing.
- Used in electronic and micro electronics devices.
- **Biomedical:** Fullerenes can be used to deliver drugs to specific cells or tissues. Used as contrast agents in imaging techniques such as MRI or ultrasound.
- **Energy storage:** Fullerenes and derivatives are used as storage devices for hydrogen
- Nonlinear optical devices.
- Environment: Sensors for detecting environmental toxins and pollutant

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Polymers

- Polymers are molecule with large molecular masses obtained by the covalent linkage of small repeated units called monomers. The repeating chemicals unit may be same or different.
- **Homo polymers**: The polymer is made one type of monomer. Eg. Poly ethylene, PVC.
- **Copolymers:** The polymer is made of different type of monomer (eg BS,ABS)

Based on the types of polymerization ,polymers can be divided into

(i) Addition polymers eg. Polythene, PVC

(ii) Condensation polymers eg. Nylon 66, bakelite

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Fire retardant polymers

- Fire retardant polymers are chemical compounds that are added to flammable polymers to make them less likely to catch fire. They can be used to: Reduce the rate of flame spread, Reduce smoke and fume emission, Reduce heat release, and Give more escape time.

  Eg. Nitrogen based polymers silicon polymers, novolac etc.

- Inorganic additives such as aluminium hydroxide, ammonium polyphosphate etc. and organophosphorus compounds can be incorporated into polymers to enhance their fire resistance.
- Halogenated polymers are also fire-retardant polymers. In the case of halogenated polymers like PVC, Poly brominated styrene etc halogen radical (Cl, Br) combines with H and OH radicals to form stable molecules like HCl and HOBr resulting in quenching of chain reaction. Among halogens Bromine is found to be more effective in quenching fire.
  Eg. Polyvinyl chloride (PVC), Poly(bromo styrene) and brominated butyl rubber.

Conducting polymers

- Organic polymers having conjugated π electron (alternate single and double bond system) can conduct electric current, such polymers are called conducting polymers

  Eg. poly acetylene, poly aniline, poly pyrrole

Classification

Conducting polymers

- Intrinsically conducting
  - polymer having conjugate π electrons
  - Doped conducting polymers
- Extrinsically conducting
  - conductive element filed polymers
  - blended conducting polymers

Intrinsically conducting polymers

- These polymers having conjugation in the back bone

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Conductive polymers having conjugated π electron

- Due to the presence of extensive conjugation in the back bone, overlapping of π electron take place and results in the formation of valence and conduction band. Which extent over entire polymer.
- Highest occupied band is valence band and lowest unoccupied band is called conduction band. These bands are separated by distance called band gap.
- When electron from Valence band is excited to conduction band either thermally or photolytically electrical conductance occurs eg. Poly acetylene and poly aniline

Doped conducting polymers

- Conducting polymers with conjugated-π electrons in their backbone can be oxidized or reduced by creating positive or negative charge. This process is referred to as Doping.

Two types of doping are

a) P-type or oxidative doping
b) n-type or reduction doping

p-doping:

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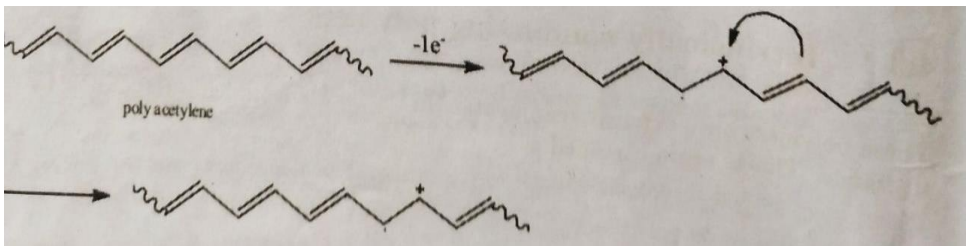
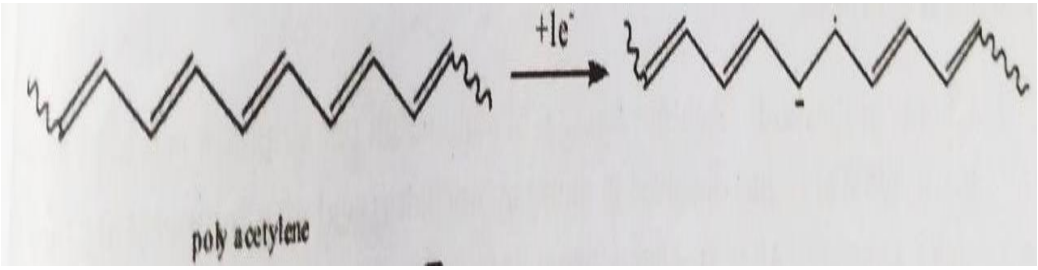
- It is done by oxidation process. In this process some electron from π bond of the conjugated double bonds are removed by Lewis acids such as $FeCl_3$ and a positive charge (hole) created. The hole (+) can move along the molecule and make the polymer conducting



n-doping: In this type of doping some electrons are introduced into the polymer having conjugated double bonds by reduction with Lewis bases like sodium naphthalide etc. Negative charge is created on the polymer chain and makes it conductive.



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Extrinsically conducting polymers

- Conducting polymers whose conductivity is due to the presence of externally added ingredients in them.
- **Conductive element filled polymer:** polymers which are filled with conducting elements such as carbon black, metallic fibers, metal powders etc. These polymers are low cost, light weight, mechanically durable and strong.

Blended conducting polymers:

- These polymers are obtained by blending conducting polymers with conventional polymers.
- These polymers possess better physical, chemical and mechanical properties.

Applications of conducting polymers

- The conducting polymers can be used as cathode and solid electrolytes in rechargeable batteries for automotive and other applications.
- It can exhibit electro chromic property and can be used to produce "smart widows and electroluminescence OLED display in TV
- Used for making analytical sensors for pH, O_2, NO_2 and glucose detection.
- In electronics it can be used in LEDS, electromagnetic shielding etc.

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Polyaniline (PANI)

- The oxidative product of aniline under acidic conditions is known as aniline black

Chemical method

- Aqueous solution of ammonium per sulphate is added slowly to acidified aniline at a temperature of 5°C with continuous stirring. After an hour a precipitate is formed, which can be removed by filtration. The polymer obtained is washed with ammonium hydroxide solution

Electrochemical method

- Platinum foil electrodes is dipped on acidified solution of aniline and a potential of 0.7 V applied, thin film of poly aniline is formed on the surface of the positive electrode.

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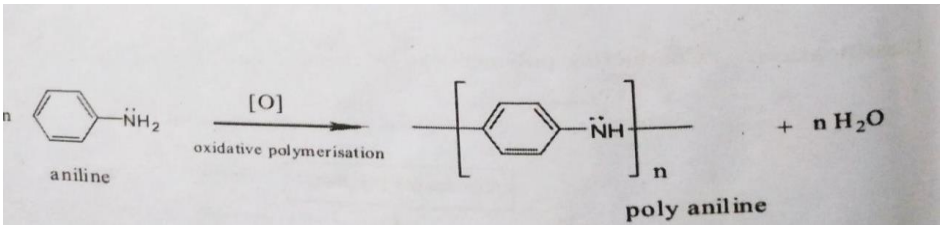

Structure: Polyaniline is found in three oxidation states in which emeraldine is conductive. Emeraldine salt of polyaniline is highly electrically conducting.

properties

- Poly aniline (PANI) is stable in air, soluble in organic solvents, chemical resistance and good conductor of electricity by suitable dopant

Applications

- As electrode material for rechargeable batteries.
- As shield in complex electronic circuits and electromagnetic radiations.
- In biosensors and solar energy absorption.

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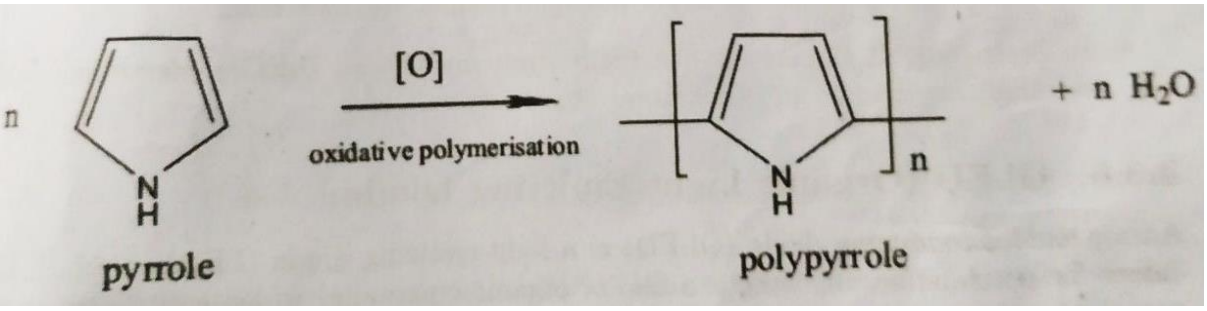
Poly pyrrole (PPy)

- It is an intrinsic conducting polymers

Chemical method: Poly pyrrole can be synthesized by oxidative polymerization. By mixing 0.1M aqueous solution of pyrrole and 0.1M of ammonium per sulphate in 1:1 ratio in a beaker and kept for 3hrs. The precipitates formed are Poly pyrrole, filtrate and dried.

Electrochemical method: By the electrolysis of pyrrole (0.06M) in aqueous acetonitrile solution using platinum electrodes and tetraethyl ammonium boron tetra fluoride as the supporting electrolyte (0.01M). The polymer will be precipitated as a blue black film on the anode.



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Properties

- PPy has excellent electrical, thermal and mechanical properties.
- It has good environmental stability and higher conductivity.
- The electrical resistance was found to decrease as temperature increases and hence conductivity increases.

Applications of Poly pyrrole (PPy)

- Poly pyrrole is used mostly in biosensors and immunosensors.
- Light weight and used in rechargeable batteries.
- It has applications in tissue engineering.
- It shows different colours in different oxidation state, hence used in electro chromic display.

OLED (Organic Light Emitting Diode)

- OLED is a light-emitting diode (LED) in which emissive electroluminescent layer is a film of organic compound which emits light in response to an electric current.
- This layer of organic semiconductor is situated between two electrodes; at least one of these electrodes is transparent.
- There are two main families of OLED: (i) those employing conducting polymers
  (ii) Those based on small molecules.

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Principle

- In conjugated polymer OLED is HOMO- LUMO transition.
- In small molecule OLED is the creation of a light-emitting electrochemical cell (LEC) on adding mobile ions to an OLED

Construction

The OLED consists of following parts

Substrate: Plastic or glass substrate, which supports OLED

Anode: Indium titanium oxide (ITO) is anode

Conductive/Hole transport layer (HTL): An organic layer that transport holes from anode eg. Poly aniline

Emissive/electron transport layer (EML/ETL): The emissive layer used in an OLED Polymer eg. Poly fluorene

Cathode: Metallic cathode such as silver copper etc.

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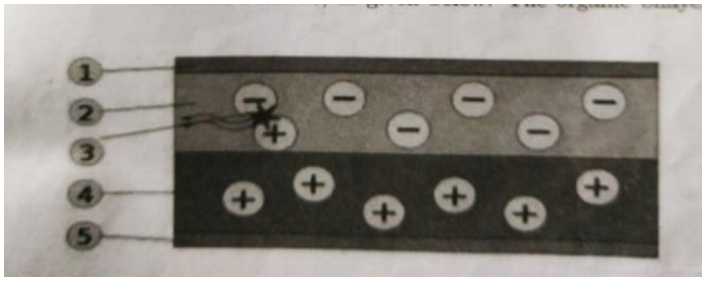

1.Cathode, 2. Emissive layer, 3. Emission of Radiation, 4. conductive layer, 5. Anode

Working

- To make an OLED light up, we simply attach a voltage (potential difference) across the anode and cathode.
- As the electricity flows, the cathode receives electrons from the power source and the anode loses them.
- An added electron makes the emissive layer negatively charged, while the conductive layer is becoming positively charged.
- Positive holes are much more mobile than negative electrons so they jump across the boundary from the conductive layer to the emissive layer.
- When a hole meets an electron enormous amount of energy is released in the form of a quantum of light or a photon.
- This process is called recombination, and it is happening many times a second, thus the OLED produces continuous light.

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Advantages of OLED

- The manufacture of OLED is highly economical and is more efficient than LCD flat panel screens.
- There is much difference in watching a high-definition TV to a OLED display. As the contrast ratio of OLED is very high (even in dark conditions), it can be watches from an angle nearly 90 degrees without any difficulty.
- No backlight is produced by this device and the power consumption is also less.
- OLED has less response time than LCD.

Dye sensitized solar cells (DSSC)

Solar cells convert light energy directly into electricity by the principle of photovoltaic effect. Photon of light knocks electrons from atoms within the semiconducting materials, creating an electric current. A dye sensitized solar cell is a photochemical cell which converts solar energy absorbed by the dye to electrical energy.

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Construction

Substrate: Glass or flexible transparent plastic in which transparent conductive oxide (TCO) is coated

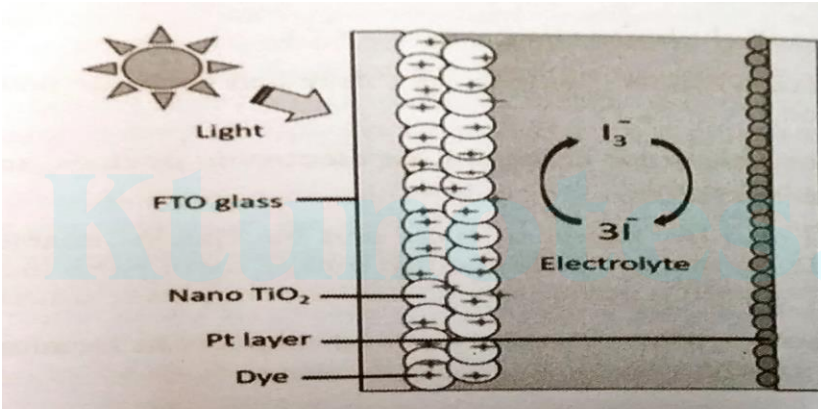
Transparent conductive oxide layer: Thin film of Indium Tin Oxide act as TCO

Semiconductor metal oxide nano particle: semiconducting metal oxide with large surface area and wide band gap are used DSSCs eg. TiO_2 , ZnO etc

Dye / Photosensitiser: surface of the nanomaterial are coated with organic or inorganic dye molecules which absorb sunlight. Dye-sensitized metal oxide semiconductor act as anode. eg Ruthenium based complex, organic dyes ,natural dyes extracted from plants etc.

Electrolyte: Redox couple iodide solution is used as electrolyte

Counter electrode: Platinum or graphite materials are used as counter electrode and it completes the circuit.



Working of Dye-sensitized solar cells

- Sunlight passes through the transparent conductive oxide layer, the metal oxide semiconductor absorb the light.
- The electron in the dye molecules get excited and are injected into the conduction band of the semiconductor. These electrons travel through external circuit and generate current.
- The oxidised dye molecules get reduced by gaining electrons from electrolyte and return back to original form.

Advantages

- The fabrication of DSSCs is simpler processes compared to conventional silicone cells.
- They are light weight which can be advantageous for application is on vehicles and pa devices.
- They are flexible in shape, and size and hence can be implemented on window, terraces etc.
- Since natural pigments and organic dyes are used, they are sustainable and eco-friendly

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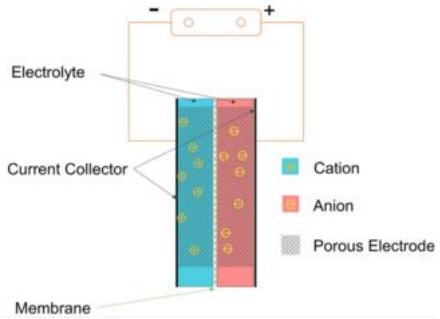
Applications

- They can be used in portable solar chargers for electronic devices
- They can be integrated into the fabric of bags and backpacks to charge electronic devices while on the travel.
- DSSC can be used to power illuminated signs and displays in locations where acces electricity is limited or costly.

Super capacitors

- A super capacitor, also known as an ultra-capacitor or double-layer capacitor, is an energy storage device having high-capacity than solid-state capacitors but with lower voltage limits.
- Super capacitors have a high power density and can release energy efficiently over a short time.
- Super capacitors can charge and discharge rapidly.



- Super capacitor generally consists of electrodes, electrolyte, separator and collector.
- It store charges at the interface between an electrode and an electrolytic solution.

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Electrodes

- Carbon-based materials are often used as super capacitor electrode materials eg. Carbon nanotubes, nanowires, metal oxide nano particles etc. Nano materials enhance the performance of the super capacitor.

Electrolyte

- Capacity of the capacitor depends mainly on the dielectric constant of the electrolyte.
- Either solid or liquid can be used as the electrolyte.
- Generally, a solvent mixed with conductive salts such as tetra alkyl ammonium or lithium salts acts as the solid electrolyte.
- Sulphuric acid, KOH solution etc can act as the liquid electrolyte.

Separator

- Electrolyte membranes will act as the separator.
- It prevents short-circuiting between the electrodes but allows the electrolyte ions to pass through.
- Polymer-based and paper- based are found to be durable and economical.

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Collector

- Electrons are collected at the collector. Carbon fibre, metal like Al, Pt, Cu etc are commonly used collectors.
- Binders, such as polyvinylidene fluoride (PVDF) and poly acrylonitrile (PAN), act as adhesives to hold the active material and current collectors together

Advantages of super capacitor

- Longer cycling time and higher service life compared to the battery.
- High efficiency, higher charging rates and high power density.
- Low resistance which enables them to produce high load currents.
- Small sizes and lightweight which makes them easily installed in small areas.
- Environment friendly.

Quantum computing technology

Quantum computing uses the principle of quantum theory. Quantum computers harness the properties such as

- superposition (quantum particles as combination of all possible states),
- Entanglement (ability of quantum particles to correlate their measurement results with each other) and
- Quantum interference (behaviour of a qubit, arising from superposition, which influences the probability of the qubit collapsing to one state or another upon measurement) for computing

In conventional computers using semiconductors, electrons represent binary states (0 or 1), but Quantum computing uses quantum bits (qubits), that are superpositions of 'spin up' and 'spin down' states, allowing them to represent all numbers between 0 and 255 simultaneously. This allows quantum computers to perform many calculations in parallel.

Materials used for Quantum Computing

- Quantum computing technology uses a variety of materials, including metals, semiconductors, and topological materials.
- Aluminum, niobium, and titanium nitride are used to create superconducting qubits.
- Silicon and other semiconductor materials are used in qubits based on electron or nuclear spins.
- Quntum dot materials like Indium Arsenide and Gallium Arsenide are used to produce photons. Photons are used as qubits in quantum communication.

Spintronics

- Spintronics or spin-electronics exploit the spin of electron in addition to charge, for data storage, processing and communication in solid devices (new generation transistors, lasers and magnetic sensors).
- Spin orientation lasts longer charge-based information, making them suitable for memory storage, magnetic sensor quantum computing.
- Spintronics make use of the magnetoresistive effects for sensing and storing information.

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- Giant Magnetoresistance (GMR) is a phenomenon where a material's electrical resistance changes dramatically when exposed to a magnetic field. This effect is widely used in magnetic storage (eg. Magnetic random-access memory) and sensing technologies.
- GMR spin valve sensor has improved the capacity of hard disc drive by hundred fold times.

Types of Spintronics

Metal based: Electron spin in metallic materials eg. MgO, Al_2O_3

Semiconductor based: Ternary alloys such as InAs and GaAs alloyed with Mn. (Co, Fe, B), etc are used as semiconductor spintronic materials due to their tunable magnetic and electronic properties.

Graphene based: Spin depended physical properties of Graphenes are also used for development of Spintronics.

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